

Non-Linear Dynamics Notes

Vyas Mokal

January 6, 2026

Contents

1		5
1.1	Books	5
1.2	Dynamics	5
1.3	1D Maps	6
1.3.1	Fixed Point	6
1.3.2	Stability of a Fixed Point	6
1.3.3	Diagrammatic way to find number of fixed points	8
1.4	Logistic Map	11

Lecture 1

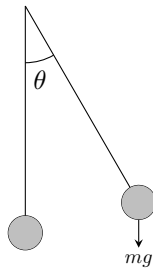
1.1 Books

- Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd ed., Westview Press, 2015.

This is Bible for the subject. It is standard, detailed and a bit old text.

1.2 Dynamics

- Study of evolution of systems
Eg. Simple Pendulum



has equation of motion

$$\ddot{\theta} - \frac{g}{l} \sin \theta = 0 \quad (1.1)$$

For small angles

$$\ddot{\theta} - \frac{g}{l} \theta = 0 \quad (1.2)$$

Which is linear. But if θ is not small then non-linear and hence included in NLD¹.

- As equation becomes Non-Linear, the solution drastically changes

¹**NLD**: Non-Linear Dynamics

- Non-Negligible Non-Linear terms are not perturbative and needs separate study
- The subject NLD has advanced after the computers became powerful enough to perform high level computation

1.3 1D Maps

- Suppose we have a sequence of numbers (real numbers mostly)

$$x_0, x_1, x_2, \dots$$

Such that

$$x_0, x_1 = f(x_0), x_2 = f(x_1), \dots \quad (1.3)$$

i.e.

$$x_{n+1} = f(x_n) \quad (1.4)$$

Then it is called an *Map*²

1.3.1 Fixed Point

- Solution of (1.3) such that

$$x^* = f(x^*) \quad (1.5)$$

then x^* ³ is called *fixed point*. Note that it will give rise to the sequence of form

$$x_0, x_1, x_2, \dots, x^*, x^*, x^*, \dots$$

1.3.2 Stability of a Fixed Point

- Suppose $x_0 = \bar{x} + \delta$.

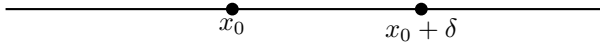
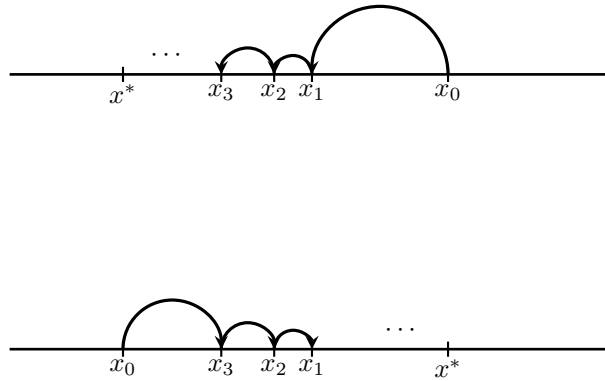


Fig. 1

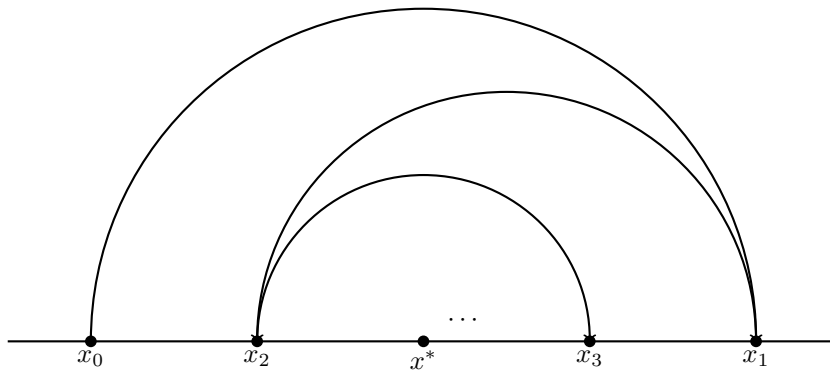
- Now if the subsequent x 's approach the fixed point x^* then the fixed point is stable and called an *attractor*

²Note here that the function f remains same

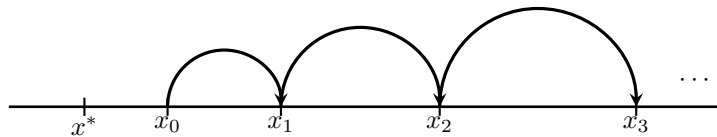
³It is widely followed conventions to label fixed point as x^* and we'll continue using it

**Fig. 2**

- Approach to x^* is Oscillatory and is still fixed point

**Fig. 3**

- If the subsequent points diverge from the fixed point (monotonically or oscillatory) then the point x^* is Unstable Fixed Point

**Fig. 4**

1.3.3 Digramatic way to find number of fixed points

Follow the steps:

1. Draw the line $y = x$
2. Draw the curve $y = f(x)$
3. No. of intersections = No. of Fixed Points⁴

Example 1. $f(x) = \sin x$

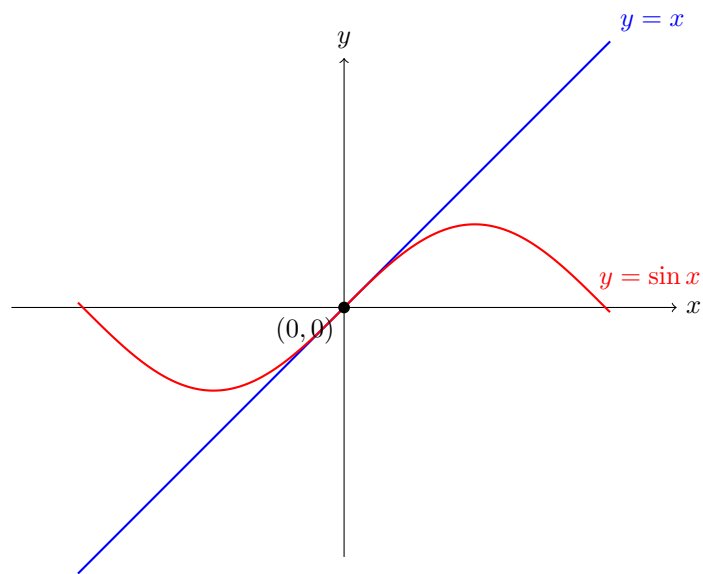


Fig. 5

There is only one fixed point at $x = 0$

Example 2. $f(x) = \cos x$

⁴This logic is clear from the definition of the fixed point as (1.3.1)

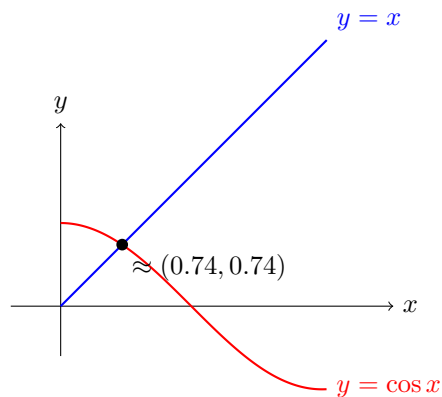


Fig. 6

There is only one fixed point at $x \approx 0.74$

Example 3. $f(x) = \frac{2x}{1+x}$

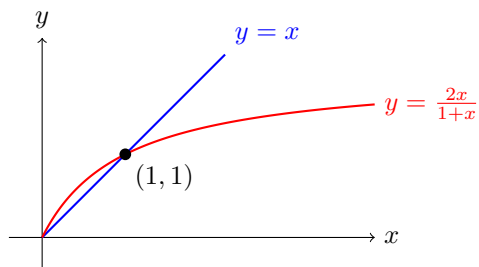


Fig. 7

There is only one fixed point at $x = 1$

- In monotonic approach
 $|x_{n+1} - x^*| < |x_n - x^*| \implies$ Stable fixed point
 $|x_{n+1} - x^*| > |x_n - x^*| \implies$ Unstable fixed point
- We define the quantity *multiplier* by

$$\mu = \frac{x_1 - x^*}{x_0 - x^*} \quad (1.6)$$

It determines whether the fixed point is stable or unstable.

$$|\mu| < 1 \implies \text{Stable} \quad (1.7)$$

$$|\mu| > 1 \implies \text{Unstable} \quad (1.8)$$

$$\mu > 0 \implies \text{Monotonic} \quad (1.9)$$

$$\mu < 0 \implies \text{Oscillatory} \quad (1.10)$$

- Now, let's analyze the fixed point analytically. Let

$$x_0 = x^* + \delta \quad (1.11)$$

Then

$$\begin{aligned} x_1 &= f(x_0) = f(x^* + \delta) \\ &= f(x^*) + \delta f'(x^*) + \frac{1}{2}\delta^2 f''(x^*) + \dots \\ &= x^* + \delta f'(x^*) + \frac{1}{2}\delta^2 f''(x^*) + \dots \end{aligned}$$

- Now linearize

$$x_1 = x^* + \delta f'(x^*) \quad (1.12)$$

Therefore,

$$f'(x^*) = \frac{x_1 - x^*}{\delta} = \frac{x_1 - x^*}{x_0 - x^*}$$

Using (1.6)

$$\mu = f'(x^*) \quad (1.13)$$

- Now in Example 1: $\mu = \cos x^* = 1$. The case $\mu = 1$ needs special analysis and we skip it for a moment
- In Example 2: $\mu = -\sin(x^*) < 0$, hence fixed point is stable and oscillatory.
- In Example 3:

$$\mu = \frac{2(1+x^*) - 2x^*}{(1+x^*)^2} = \frac{2}{(1+x^*)^2}$$

At $x^* = 0$; $\mu = 2 \implies$ monotonically unstable

At $x^* = 1$; $\mu = 1/2 \implies$ monotonically stable

- Calculating Stability form multiplier is only valid when Linearization is valid.
- We now adopt Diagramatic Approach: *Cobwebb Diagrams*
 1. Draw curve $y = x$
 2. Draw $y = f(x)$
 3. Draw line from x_0 to $f(x)$
 4. Draw horizontal line to $y = x$ and that point of intersection gives x_1

1.4 Logistic Map

We define *Logistic Map*⁵⁶ by

$$f(x) = rx(1 - x) \quad r > 0 \quad (1.14)$$

- Now let's find the fixed points

⁵ $f(x)$ is quadratic hence included in NLD

⁶This equations appears in many areas: Engineernig, Population Dynamics, etc.